

popquiz Work-KE, Hookes law

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:38pm

Quiz Instructions

1933 movie. Amazing for the time.



Question 1 25 pts



King Kong had a mass of about 3500kg in the first movie. The Empire state building is 440m high. If king kong falls and we neglect air resistance, what is its final speed JUST before it hits the floor (initial speed is 0) ?

You can use work-kinetic theorem : work done by gravity = change in kinetic energy.

and work done by gravity = weight x height.

Note: you can also use the last equation of kinematics $V^2 = V_i^2 - 2 a h$ Its the same thing ! Dah

Take $g = 10\text{m/s}^2$ (its free-fall the mass of king-kong does not matter while falling. Its too heavy for air resistance to build enough. Its the decelarion that will kill him)

☐

about 35m/s (70mph)

☐

about 25m/s (50mph)

☐

about 45m/s (90mph)

☐

about 95m/s (210mph)



Question 2 25 pts

Building from the previous question. King Kong reaches the floor with the previous speed. It is acted upon by the normal force and its weight. It stops in a short time. Because it is slowing down then stops the normal force N is ? (in magnitude). Remember velocity is down but acceleration is up because it is decelerating. The net force needs to have the same direction as the acceleration. Dah

hint: N is the apparent weight. Imagine King Kong falling on a bathroom scale). It is also the force experienced by poor King King OUCH

What is true

☐

$N = \text{weight}$

☐

$N < \text{weight}$

☐

$N > \text{weight}$

☐

0



Question 3 25 pts

Still building from the previous questions.

When King Kong touches the floor, it is acted by the Normal force and the weight and its speed decreases from the speed V found in Q1 to 0 (it stops)

The work- KE theorem says: Work done by weight + work done by normal = change in KE.

What is the right equation to find the Normal force (force acting on King Kong and killing it). Suppose King Kong goes into the asphalt by a distance d (a few cm)

Normal and weight used in the equations are the magnitude of the forces,

work is negative if the force tries to slow down the system (which force is slowing it down !!! not the weight)

it's not the fall that kills , it's the landing Dah

negative work is when a force is slowing down the system. positive work is when a force is speeding up the system.

☐

weight $\times d$ + Normal $\times d = 0.5 \text{ m (} V^2)$

☐

weight $\times d$ - Normal $\times d = 0.5 \text{ m (} V^2)$

☐

weight $\times d$ - Normal $\times d = 0.5 \text{ m (0 - } V^2)$

☐

weight $\times d$ + Normal $\times d = 0.5 \text{ m (0 - } V^2)$



Question 4 25 pts

Building on the previous question

What is the force experience by king king. Suppose $d = 0.01\text{m}$ (1 cm). Say the ground has some yield

Make sure you got the previous question right. The normal force is doing negative work. the weight is doing positive work.

☐

about 2 KN (divide by 50 get pounds)

☐

about 2 billion N (divide by 50 get pounds)

☐

about 2 trillion N (divide by 50 get pounds)

☐

about 2 million N (divide by 50 get pounds)

Not saved

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